

Business Case:

1) Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

1) Data type of all columns in the "customers" table.

Filter Enter property name or value ?

☐	Field name	Type	Mode	Key	Collation	Default Value	Policy Tags ?	Data Policies	Description
☐	customer_id	STRING	NULLABLE	-	-	-	-	-	-
☐	customer_unique_id	STRING	NULLABLE	-	-	-	-	-	-
☐	customer_zip_code_prefix	INTEGER	NULLABLE	-	-	-	-	-	-
☐	customer_city	STRING	NULLABLE	-	-	-	-	-	-
☐	customer_state	STRING	NULLABLE	-	-	-	-	-	-

Insights:

- 4 STRINGS and 1 INTEGER DATA types
- Clarity on states and cities gives marketing to company.

2) Get the time range between which the orders were placed.

```
select
min(order_purchase_timestamp)as first_order,
max(order_purchase_timestamp)as last_order
from `Business_Case.orders`
```

output:

Query results

Job information		Results	Visualization	JSON	Execution details	Execution graph
Row	first_order	last_order				
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC				

Insights:

- Time range between first order to last order is 772days,20 hours,14,minutes and 59 seconds

Suggestions:

- Improve Logistics & Freight
- Customer Retention Strategy
- Customer Reviews & Experience

3)Count the Cities & States of customers who ordered during the given period.

```
SELECT
COUNT(DISTINCT c.customer_city) AS num_cities,
COUNT(DISTINCT c.customer_state) AS num_states
FROM
`Business_Case.orders` o
JOIN
`Business_Case.customers` c
ON
o.customer_id = c.customer_id;
```

output:

Query results

Job information		Results	Visualization	JSON	Execution details	Execution graph
Row	num_cities	num_states				
1	4119	27				

Insights:

- In a given period there were 4119 cities and 27 states of customers were there.

◆ Suggestions :

- Serving 27 states & 4,119 cities shows broad coverage.
- However, logistics challenges (long freight times, costs) may impact customer satisfaction in remote areas
- A span of 772 days between first & last orders shows steady engagement.
- Useful for studying seasonal trends (holidays, events like Black Friday, Christmas)

◆ Recommendations:

- Partner with local courier networks for last-mile delivery.
- Offer loyalty discounts for frequent customers.
- Provide free/discounted shipping for smaller states to encourage adoption.
- Personalize marketing based on city/state-level purchase behavior.
- Explore subscription models (free shipping, priority delivery, exclusive discounts).

2) In-depth Exploration:

1) Is there a growing trend in the no. of orders placed over the past years?

```
SELECT
  EXTRACT(YEAR FROM TIMESTAMP(order_purchase_timestamp)) AS order_year,
  COUNT(order_id) AS num_orders
FROM
  `Business_Case.orders`
GROUP BY
  order_year
ORDER BY
  order_year;
```

Output:

Query results

Job information		Results	Visualization	JSON	Execution details	Execution graph
Row	order_year ▼	num_orders ▼				
1	2016	329				
2	2017	45101				
3	2018	54011				

Insights:

 **Analysis & Observations**

1. Rapid Growth:

- From 329 (2016) → 45,101 (2017) = ~13,600% growth 🚀
- From 2017 → 2018, growth slowed to ~19.8%.
👉 Business reached a *scaling* stage after explosive initial adoption

✅ Suggestions:

- Analyze order history → recommend products, offer tailored discounts.
- Use RFM analysis (Recency, Frequency, Monetary) to segment customers.
- Introduce automated inventory management to match higher order volumes.

📌 Recommendations:

- Strengthen customer support and delivery speed (build trust).
- Build brand recognition (content marketing, influencer tie-ups).
- Run retention-focused campaigns (email marketing, loyalty rewards).

2) Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

SELECT

```
EXTRACT(YEAR FROM TIMESTAMP(order_purchase_timestamp)) AS order_year,  
EXTRACT(MONTH FROM TIMESTAMP(order_purchase_timestamp)) AS order_month,  
COUNT(order_id) AS num_orders
```

FROM

```
`Business_Case.orders`
```

GROUP BY

```
order_year, order_month
```

ORDER BY

```
order_year, order_month;
```

Output:

Query results

Job information	Results	Visualization	JSON	Execut
Row	order_year ▼	order_month ▼	num_orders ▼	
1	2016	9	4	
2	2016	10	324	
3	2016	12	1	
4	2017	1	800	
5	2017	2	1780	
6	2017	3	2682	

Job history

Job information	Results	Visualization	JSON	Exec
Row	order_year ▼	order_month ▼	num_orders ▼	
7	2017	4	2404	
8	2017	5	3700	
9	2017	6	3245	
10	2017	7	4026	
11	2017	8	4331	
12	2017	9	4285	

Job history

Insights:

- Strong Growth Phase: From 2016 (a few hundred orders) → 2017 (thousands monthly). Target Brazil was in its early scaling stage.

Seasonality Present:

- Peaks: May, Jul, Aug (likely tied to Mother's Day in May, mid-year sales).

◆ Recommendations:

- Focus campaigns around May, July, August, and Nov-Dec (seasonal high-demand months).
- Forecast demand spikes (like May & Aug) to avoid stockouts.
- Scaleup manpower and products as per demand.

3) During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)

- 0-6 hrs : Dawn
- 7-12 hrs : Mornings
- 13-18 hrs : Afternoon
- 19-23 hrs : Night

```
SELECT
CASE
    WHEN EXTRACT(HOUR FROM TIMESTAMP(o.order_purchase_timestamp)) BETWEEN 0 AND 6 THEN
'Dawn'
    WHEN EXTRACT(HOUR FROM TIMESTAMP(o.order_purchase_timestamp)) BETWEEN 7 AND 12 THEN
'Morning'
    WHEN EXTRACT(HOUR FROM TIMESTAMP(o.order_purchase_timestamp)) BETWEEN 13 AND 18
THEN 'Afternoon'
    WHEN EXTRACT(HOUR FROM TIMESTAMP(o.order_purchase_timestamp)) BETWEEN 19 AND 23
THEN 'Night'
END AS time_of_day,
COUNT(o.order_id) AS num_orders
FROM
`Business_Case.orders` o
GROUP BY
```

```
time_of_day
ORDER BY
num_orders DESC;
```

output:

Query results

Job information	Results	Visualization	JSON
row	time_of_day ▼	num_orders ▼	
1	Afternoon	38135	
2	Night	28331	
3	Morning	27733	
4	Dawn	5242	

Insights:

- Afternoon (13–18 hrs): 38,135 orders → Highest activity with ≈36% of total orders
- Night (19–23 hrs): 28,331 orders → Strong 2nd peak
- Morning (7–12 hrs): 27,733 orders → Close to Night, still significant
- Dawn (0–6 hrs): 5,242 orders → Very low activity with ≈5%

Recommendations:

- Peak focus: Run ads & promotions in Afternoon (12–18 hrs) and Night (19–23 hrs).
- Ensure delivery partners are ready for high evening spikes.
- Strengthen server performance and app responsiveness between 12–23 hrs.

3) Evolution of E-commerce orders in the Brazil region:

1) Get the month on month no. of orders placed in each state.

```
SELECT
  EXTRACT(YEAR FROM TIMESTAMP(o.order_purchase_timestamp)) AS order_year,
  EXTRACT(MONTH FROM TIMESTAMP(o.order_purchase_timestamp)) AS order_month,
  c.customer_state,
  COUNT(o.order_id) AS num_orders
FROM
  `Business_Case.orders` o
JOIN
  `Business_Case.customers` c
ON
  o.customer_id = c.customer_id
GROUP BY
  order_year, order_month, c.customer_state
ORDER BY
  c.customer_state;
```

output:

Query results

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	order_year ▼	order_month ▼	customer_state ▼	num_orders ▼	
1	2016	9	RR	1	
2	2016	9	RS	1	
3	2016	9	SP	2	
4	2016	10	AL	2	
5	2016	10	BA	4	
6	2016	10	CE	8	

Results per page

Query results

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	order_year ▼	order_month ▼	customer_state ▼	num_orders ▼	
7	2016	10	DF	6	
8	2016	10	ES	4	
9	2016	10	GO	9	
10	2016	10	MA	4	
11	2016	10	MG	40	
12	2016	10	MT	3	

Results per page

Insights:

- Smaller states (e.g., RR, AL, AC) show very few orders — possibly due to logistics or digital adoption issues.
- You'll likely notice **peaks around November/December** in many states

Recommendations:

- Increase marketing spend, loyalty programs, and faster delivery options in states like SP, RJ, MG.
- Partner with 3PLs (third-party logistics) or provide free/discounted shipping to encourage adoption.

2)How are the customers distributed across all the states?

```
SELECT
customer_state,
COUNT(DISTINCT customer_id) AS num_customers
FROM
`Business_Case.customers`
GROUP BY
customer_state
ORDER BY
customer_state,num_customers DESC;
```

Output:

Query results

Job information		Results	Visualization	JSON	Execution
Row	customer_state ▼	num_customers ▼			
1	AC	81			
2	AL	413			
3	AM	148			
4	AP	68			
5	BA	3380			
6	CE	1336			

Query results

Job information		Results	Visualization	JSON
Row	customer_state ▼	num_customers ▼		
7	DF	2140		
8	ES	2033		
9	GO	2020		
10	MA	747		
11	MG	11635		
12	MS	715		

Insights:

- Need to much focus on states like AC,AP
- GIVING view on states and distributed customers

Recommendations :

- Need more campaigns in high density states
- Need collaborations to reach more masses

4)Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

1)Get the % increase in the cost of orders from year 2017 to 2018 *(include months between Jan to Aug only)*.

You can use the “payment_value” column in the payments table to get the cost of orders.

```

SELECT
    ((SUM(CASE WHEN EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2018
    THEN p.payment_value END) -
        SUM(CASE WHEN EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2017
    THEN p.payment_value END)
    ) / SUM(CASE WHEN EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2017
    THEN p.payment_value END)
    ) * 100 AS pct_increase
FROM
    `Business_Case.orders` o
    JOIN `Business_Case.payments` p
    ON o.order_id = p.order_id

WHERE
    EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 8
    AND EXTRACT(YEAR FROM o.order_purchase_timestamp) IN (2017, 2018);

```

output:

Query results

Job information		Results	Visualization	JSON	Execution
Row	pct_increase ▼				
1	136.9768716466...				

Insights:

- Strong in revenue has took place.
- Customer Base Expansion has improved

Recommendations:

- Identify Key Drivers of Growth like(Break down by state, city, or payment method to see which area contributed to make surge)
- Forecast Future Demand
- Customer Retention(through offers)

2)Calculate the Total & Average value of order price for each state.

```

SELECT
    c.customer_state,
    ROUND(SUM(oi.price),2) AS total_order_value,
    ROUND(AVG(oi.price),2) AS avg_order_value
FROM
    `Business_Case.order_items` oi
JOIN
    `Business_Case.orders` o
    ON oi.order_id = o.order_id
JOIN
    `Business_Case.customers` c
    ON o.customer_id = c.customer_id
GROUP BY
    c.customer_state
ORDER BY
    c.customer_state ,total_order_value,avg_order_value DESC;

```

output:
Query results

Job information		Results	Visualization	JSON	Execution details	Exe
Row	customer_state ▼	total_order_value ▼	avg_order_value ▼			
1	AC	15982.95	173.73			
2	AL	80314.81	180.89			
3	AM	22356.84	135.5			
4	AP	13474.3	164.32			
5	BA	511349.99	134.6			
6	CE	227254.71	153.76			

Query results

Job information	Results	Visualization	JSON	Execution details
w	customer_state ▼	total_order_value ▼	avg_order_value ▼	
7	DF	302603.94	125.77	
8	ES	275037.31	121.91	
9	GO	294591.95	126.27	
10	MA	119648.22	145.2	
11	MG	1585308.03	120.75	
12	MS	116812.64	142.63	

Insights:

- Calculate the Total & Average value of order price for each state.
- **CE** has both a **solid total value (227,254.71)** and a **decent average order value (153.76)** → good balance of customer volume and spend.

Recommendations:

- **Focus on upselling and cross-selling to increase average basket size.**
- **Run targeted campaigns (discounts, free shipping, local-language ads) to boost engagement.**

3) Calculate the Total & Average value of order freight for each state.

```
SELECT
c.customer_state,
SUM(oi.freight_value) AS total_freight_value,
ROUND(AVG(oi.freight_value), 2) AS avg_freight_value
FROM
`Business_Case.orders` o
JOIN
`Business_Case.customers` c
ON
o.customer_id = c.customer_id
JOIN
`Business_Case.order_items` oi
```

ON

o.order_id = oi.order_id

GROUP BY

c.customer_state

ORDER BY

c.customer_state, total_freight_value, avg_freight_value DESC;

output:

Query results

Job information Results Visualization JSON Execution details				
Row	// customer_state ▼	// total_freight_value ▼	avg_freight_value ▼	//
1	AC	3686.750000000...	40.07	
2	AL	15914.589999999...	35.84	
3	AM	5478.890000000...	33.21	
4	AP	2788.499999999...	34.01	
5	BA	100156.6800000...	26.36	
6	CE	48351.590000000...	32.71	

Query results

Job information Results Visualization JSON Execution details				
Row	// customer_state ▼	// total_freight_value ▼	avg_freight_value ▼	//
7	DF	50625.499999999...	21.04	
8	ES	49764.600000000...	22.06	
9	GO	53114.980000000...	22.77	
10	MA	31523.769999999...	38.26	
11	MG	270853.4600000...	20.63	
12	MS	19144.03	23.37	

Insights and impact on economy:

- High volumes of shipments → Indicates strong consumer demand and trade flow.

- Potential for economic growth if e-commerce and logistics investments increase.

Recommendations:

- Focus on optimizing logistics: Since order volume is high, explore bulk shipping discounts, regional warehouses, and optimized delivery routes.
- Consider setting up micro-fulfillment centers or local partnerships to reduce per-order freight costs.

5) Analysis based on sales, freight and delivery time.

1) Find the no. of days taken to deliver each order from the order's purchase date as delivery time.

Also, calculate the difference (in days) between the estimated & actual delivery date of an order.

Do this in a single query.

You can calculate the delivery time and the difference between the estimated & actual delivery date using the given formula:

- **time_to_deliver** = order_delivered_customer_date - order_purchase_timestamp.
- **diff_estimated_delivery** = order_estimated_delivery_date - order_delivered_customer_date

SELECT

o.order_id,

DATE_DIFF(o.order_delivered_customer_date,o.order_purchase_timestamp, DAY) AS

time_to_deliver,

DATE_DIFF(o.order_estimated_delivery_date,o.order_delivered_customer_date, DAY) AS

diff_estimated_delivery

FROM

`Business\_Case.orders` o

order by 1,2,3;

Output:

Query results

Job information					Results	Visualization	JSON	Execution details
Row	order_id	time_to_deliver	diff_estimated_de...					
1	00010242fe8c5a6d1ba2dd792c...	7	8					
2	00018f77f2f0320c557190d7a1...	16	2					
3	000229ec398224ef6ca0657da4...	7	13					
4	00024acbcd0a6daa1e931b038...	6	5					
5	00042b26cf59d7ce69dfabb4e5...	25	15					
6	00048cc3ae777c65dbb7d2a06...	6	14					

Query results

Job information					Results	Visualization	JSON	Execution details
Row	order_id	time_to_deliver	diff_estimated_de...					
7	00054e8431b9d7675808bcb81...	8	16					
8	000576fe39319847cbb9d288c5...	5	15					
9	0005a1a1728c9d785b8e2b08b...	9	0					
10	0005f50442cb953dcd1d21e1fb...	2	18					
11	00061f2a7bc09da83e415a52dc...	4	10					
12	00063b381e2406b52ad429470...	10	0					

Insights:

- Higher-value orders are spread across states .
- Many orders are being delivered earlier than estimated.

Recommendations:

- Focus on regions with 25+ days average delivery. Partner with local carriers or build regional warehouses.

- Introduce minimum order value for free shipping to maintain profitability.
- Adjust estimates to be realistic but competitive with industry benchmarks.

2) Find out the top 5 states with the highest & lowest average freight value.

top 5 states with the highest:

```
SELECT
    c.customer_state,
    ROUND(AVG(oi.freight_value), 2) AS avg_freight_value
FROM
    `Business_Case.order_items` oi
JOIN
    `Business_Case.orders` o
    ON oi.order_id = o.order_id
JOIN
    `Business_Case.customers` c
    ON o.customer_id = c.customer_id
GROUP BY
    c.customer_state
ORDER BY
    avg_freight_value DESC
LIMIT 5;
```

Top 5 states lowest average freight value:

```
SELECT
    c.customer_state,
    ROUND(AVG(oi.freight_value), 2) AS avg_freight_value
FROM
    `Business_Case.order_items` oi
JOIN
    `Business_Case.orders` o
    ON oi.order_id = o.order_id
JOIN
    `Business_Case.customers` c
    ON o.customer_id = c.customer_id
GROUP BY
```

```
c.customer_state
ORDER BY
  Avg_freight_value asc
LIMIT 5;
```

Output:

top 5 states with the highest:

Query results

Job information		Results	Visualization	JSON
Row	customer_state ▼	avg_freight_value ▼		
1	RR	42.98		
2	PB	42.72		
3	RO	41.07		
4	AC	40.07		
5	PI	39.15		

Top 5 states lowest average freight value:

Query results

Job information	Results	Visualization	JSON	Execution details
row	customer_state ▼	avg_freight_value ▼		
1	SP	15.15		
2	PR	20.53		
3	MG	20.63		
4	RJ	20.96		
5	DF	21.04		

Insights:

- Top 5 highest and lowest avg freight value

Recommendations:

- Freight Cost Management.
- Customer Segmentation by State.

3) Find out the top 5 states with the highest & lowest average delivery time.

Top 5 states with highest average delivery time:

SELECT

```
c.customer_state,  
avg(timestamp_diff(o.order_delivered_customer_date,  
o.order_purchase_timestamp,day)) AS avg_delivery_days
```

FROM

```
`Business_Case.orders` o
```

JOIN

```
`Business_Case.customers` c  
ON o.customer_id = c.customer_id
```

GROUP BY

```
c.customer_state
```

ORDER BY

```
2 desc  
LIMIT 5;
```

Top 5 states with lowest average delivery time:

```
SELECT  
    c.customer_state,  
    avg(timestamp_diff(o.order_delivered_customer_date,  
o.order_purchase_timestamp,day)) AS avg_delivery_days  
FROM  
    `Business_Case.orders` o  
JOIN  
    `Business_Case.customers` c  
    ON o.customer_id = c.customer_id  
GROUP BY  
    c.customer_state  
ORDER BY  
    2 asc  
LIMIT 5;
```

Output:

Top 5 states with highest average delivery time:

Query results

Job information		Results	Visualization	JSON
row	customer_state ▼	avg_delivery_days ▼		
1	RR	28.97560975609...		
2	AP	26.73134328358...		
3	AM	25.98620689655...		
4	AL	24.04030226700...		
5	PA	23.31606765327...		

Top 5 states with lowest average delivery time:

Query results

Job information		Results	Visualization	JSON
row	customer_state ▼	avg_delivery_days ▼		
1	SP	8.298061489072...		
2	PR	11.52671135486...		
3	MG	11.54381329810...		
4	DF	12.50913461538...		
5	SC	14.47956019171...		

insights:

- Top 5 states with highest and lowest avg delivery time.

Recommendations:

- Improve Last-Mile Delivery
- Forecast demand in high-delay states and pre-stock popular products locally.
- Collaborate with state authorities for better transport infrastructure in under-served areas.

4))Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

You can use the difference between the averages of actual & estimated delivery date to figure out how fast the delivery was for each state.

Select

```
c.customer_state ,
avg(date_diff(order_delivered_customer_date, order_purchase_timestamp,
day)) as avg_actual_days,
avg(date_diff(order_estimated_delivery_date, order_purchase_timestamp,
day)) as avg_estimated_days,
(avg(date_diff(order_estimated_delivery_date, order_purchase_timestamp,
day)) -
avg(date_diff(order_delivered_customer_date, order_purchase_timestamp,
day))) as delivery_faster_by_days
from `Business_Case.orders` o join `Business_Case.customers` c
on o.customer_id=c.customer_id
where order_delivered_customer_date is not null and
order_estimated_delivery_date is not null
group by 1
order by 4 desc
limit 5
```

Output:

Query results

Job information		Results	Visualization	JSON	Execution details	Execution gra
w	customer_state ▼	avg_actual_days ▼	avg_estimated_d...	delivery_faster_by...		
1	AC	20.63749999999...	40.72499999999...	20.0875		
2	RO	18.91358024691...	38.38683127572...	19.47325102880...		
3	AP	26.73134328358...	45.86567164179...	19.13432835820...		
4	AM	25.98620689655...	44.9241379310345	18.93793103448...		
5	RR	28.97560975609...	45.63414634146...	16.65853658536...		

Insights:

- AC, RO, AP, AM, and RR might have stronger logistics networks, faster last-mile partners, or warehouses located closer to customers.

Recommendations:

- i) Refine delivery estimates.
- ii) Optimize inventory & warehouse planning.

6) Analysis based on the payments:

1) Find the month on month no. of orders placed using different payment types.

SELECT

```
extract(year from o.order_purchase_timestamp) as year,  
extract(month from o.order_purchase_timestamp) as month,  
p.payment_type,  
COUNT(o.order_id) AS total_orders
```

FROM

```
`Business_Case.orders` o
```

JOIN

```
`Business_Case.payments` p
```

ON

```
o.order_id = p.order_id
```

GROUP BY

```
1,2,3
```

ORDER BY

```
1,2
```


output:

Query results

Job information	Results	Visualization	JSON	Execution details	Execution
row	year	month	payment_type	total_orders	
1	2016	9	credit_card	3	
2	2016	10	voucher	23	
3	2016	10	credit_card	254	
4	2016	10	UPI	63	
5	2016	10	debit_card	2	
6	2016	12	credit_card	1	
7	2017	1	credit_card	583	
8	2017	1	UPI	197	
9	2017	1	debit_card	9	
10	2017	1	voucher	61	

Insights:

- This shows Credit Card is Dominant,UPI is Emerging,Voucher Usage is Sporadic,Debit Card is Low ,mostly customers trusting digital payments.

Recommendations:

- Offer reward points, cashback, or EMI with banks since credit card is already dominant.
- Revisit Debit Card Strategy ,cause its usage is weak.

2)Find the no. of orders placed on the basis of the payment installments that have been paid.

```
SELECT
    p.payment_installments,
    COUNT(DISTINCT p.order_id) AS total_orders
FROM
    `Business_Case.payments` p
GROUP BY
    1
ORDER BY 1,2 ;
```

OUTPUT:

Query results

Job information					Results	Visualization	JSON	Execution details	Ex
Row	payment_installm...	total_orders							
1	0	2							
2	1	49060							
3	2	12389							
4	3	10443							
5	4	7088							
6	5	5234							
7	6	3916							
8	7	1623							
9	8	4253							
10	9	644							
11	10	5315							

Insights:

- Majority use single payment

Recommendations:

- Promote single payment discounts
- Investigate long-term installment patterns(like long term EMI OPTIONS)
- Customer segmentation strategy(LIKE High-value buyers,Budget-sensitive buyers)